R15

Code No: 123AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, April/May - 2023
PROBABILITY AND STATISTICS
(Common to ME, CSE, IT, MCT, AME, MIE)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) Is the function defined by $f(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{18}(2x+3), & 2 \le x \le 4 \\ 0, & x > 4 \end{cases}$ a probability density function? [2]

b) Calculate the expectation and variance of X, if the probability distribution of the random variable X is given by [3]

X	-1	0	1	2	3
f	0.3	0.1	0.1	0.3	0.2

c) Define Marginal probability mass function and Marginal probability density function. [2]

d) Joint distribution of X and Y is given by $f(x,y) = 4xye^{x^2+y^2}; x \ge 0, y \ge 0$. Test whether X and Y are independent. [3]

e) Define population, Sampling, Statistic and parameter. [2]

f) A coin was tossed 960 times and returned heads 183 times. Test the hypothesis that a coin is unbiased. Use a 0.05 level of significance. [3]

g) Define (M/M/1) system. [2]

h) Show that average number of units in a M/M/1 system is equal to $\frac{\rho}{1-\rho}$. [3]

i) Which of the stochastic matrices are regular [2]

I) $A = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix}$ II) $B = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$

j) Define classification of possible states. [3]

PART - B

(50 Marks)

2.a) If the probability that a man aged 60 will live to be 70 is 0.65, what is the probability that out of 10 men, now 60, at least 7 will live to be 70?

b) Find the mean and standard deviation of a normal distribution in which 31% of items are under 45 and 8% are over 64. [5+5]

OR

3.a) A manufacturer knows that the condensers he makes contain on average 1% defective. He packs them in boxes of 100. What is the probability that a box picked at random will contain 3 or more faulty condensers?

b) In a sample of 1000 cases, the mean of a certain test is 14and standard deviation is 2.5. Assuming the distribution to be normal, find (i) how many students score between 12 and 15 (ii) how many score above 18 (iii) how many score below 18. [5+5]

4. A computer while calculating correlation coefficient between two variables X and Y from 25 pairs of observations obtained the following results:
$n = 25,\ \sum X = 125, \sum X^2 = 650, \sum Y = 100,\ \sum Y^2 = 460,\ \sum XY = 508.$
If was, however, later discovered at the time of checking that he had copied down two pairs as

X	Y
6	14
8	6

while the correct values were

X	Y
8	12
6	8

Obtain the correct value of correlation coefficient. [10]

OR

5. Ten competitors in a beauty contest are ranked by three judges as follows:

Judges	1	2	3	4	5	6	7	8	9	10
Ranks by A	6	5	3	10	2	4	9	7	8	1
Ranks by B	5	8	4	7	10	2	1	6	9	3
Ranks by C	4	9	8	1	2	3	10	5	7	6

Discuss which pair of judges has the nearest approach to common tastes of beauty. [10]

6.a) A cigarette manufacturing firm claims that its brand A line of cigarettes outsells its brand B by 8%. If it is found that 42 out of a sample of 200 smokers prefer brand A and 18 out of another sample of 100 smokers prefer brand B, test whether the 8% difference is a valid claim.

b) The nicotine contents in milligrams into samples of tobacco were found to be as follows:

Sample A	24	27	26	21	25	---
Sample B	27	30	28	31	22	36

Can it be said that two samples came from same normal population. Test at 5% level of significance. [5+5]

OR

7.a) A group of 5 patients treated with medicine A weight 42, 39, 48, 60 and 41 kgs. Second group of 7 patients from the same hospital treated with medicine B weight 38, 42, 56, 64, 68, 69 and 62 kgs. Do you agree with the claim that medicine B increases the weight significantly?

b) The number of automobile accidents per week in a certain community are as follows: 12, 8, 20, 2, 14, 10, 15, 6, 9, 4 are these frequencies in agreement with the belief that accident conditions were the same during this 10-week period. [5+5]

8. A self service canteen employs one cashier at its counter. 8 customers arrive per every 10 minutes on an average. The cashier can serve on an average one per minute. Assuming that the arrivals are Poisson and the service time distribution is exponential, determine: (a) average number of customers in the system: (b) the average queue length: (c) average time a customer spends in the system: (d) average waiting time of each customer. [10]

OR

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9. Consider an example from a maintenance shop the inter arrival time at tool crib are exponential with an average time of 10 minutes the length of the service time (amount of time taken by the tool crib operator to meet the needs of the maintenance man) is assumed to be exponentially distributed with mean 6 minutes. Find (a) The probability that a person arriving at the booth will have to wait. (b) average length for the queue that forms and the average time that an operator spends in the queue system. (c) the manager of the shop will install a second booth when an arrival would have to wait 10 mins or more for the service by how much was the rate of arrival be increased in order to justify the second booth. (d) the probability that an arrival will have to wait for more than 12 minutes for service and to obtain his tools. [10]

10. Three boys A, B, C are throwing a ball to each other, A always throws the ball to B and B always throws the ball to C but C just as likely to B as to A. Show that the process is Markov. Find the transition matrix and classify the states. [10]

OR

11. A training process is considered as a two state Markov chain. If it rains, it is considered to be in state 0 and doesn't rain, the chain is in the state of 1. The transition probability of the Markov chain is defined by $P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$. Find the probability that it will rain for 3 days from today assuming that it is raining today. Assume that the mutual probabilities of state 0 or state 1 as 0.4 and 0.6 respectively. [10]

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